



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

CHEMISTRY
JC2 Preliminary Exam
Paper 1(MCQ)

8872/01
23 August 2011
50 min

Additional Materials: Data Booklet
 Multiple Choice Answer

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice answer sheet given,
Write your name, NRIC, class in the spaces provided.
Shade correctly your class and NRIC/FIN
Eg. If your NRIC is S9206660Z, shade **S9206660Z**

There are **thirty** questions in this paper. Answer **all** questions.
For each question there are four possible answers **A, B, C** and **D**.
Choose the one you consider correct and record your choice in **soft pencil** on the separate answer sheet.
Each correct answer will score one mark.
A mark will not be deducted for a wrong answer.

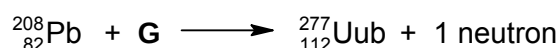
You are advised to fill in the OMS Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

This document consist of **11** printed pages and **1** blank page

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider to be correct.

- Which of the following contains the greatest amount of particles at s.t.p?
 - 600 g of I_2 (s)
 - 50 cm^3 of H_2O (l)
 - 50 dm^3 of HCl (g)
 - 550 cm^3 of 2.5 $mol\ dm^{-3}$ CH_3COOH (aq)
- A 50.0 cm^3 sample of an iron (II) solution required 34.60 cm^3 of 0.100 $mol\ dm^{-3}$ potassium dichromate (VI) solution for complete reaction.
Calculate the mass of iron (II) ions in 1 dm^3 of the iron (II) solution.
 - 0.42 g
 - 1.16 g
 - 3.86 g
 - 23.2 g
- Which electronic configuration represents an atom of an element, **E** that forms a simple ion, E^{2-} ?
 - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$
 - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
 - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$
 - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$
- In 1996, Professor Hoffman and a group of scientists discovered a new element known as Ununbium, Uub. An atom of Uub can be formed by the fusion of a lead nucleus with the nucleus of an isotope of element **G** as follows:



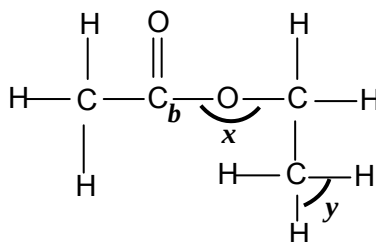
What is **G**?

- Cu
 - Ga
 - Ge
 - Zn
- Which of the species in their gaseous state shown below will be deflected by the largest angle in an electric field?
 - ${}^{32}_{16}S^{2-}$
 - ${}^4He^{2+}$
 - ${}^{16}_8O^{2-}$
 - ${}^1H^+$

6. Which of the following sets consists of three compounds with different types of structure?

- A $AlCl_3$, Cu_2O , $SiCl_4$
- B $BeCl_2$, Na_2O , SiO_2
- C AlF_3 , $MgCl_2$, BN
- D BeO , SO_2 , BH_3

7. Which one of the following options about the following ester is correct?



	Type of hybridisation of C_b	Value of bond angle x	Value of bond angle y
A	sp	180°	109°
B	sp^2	104°	109°
C	sp^2	180°	90°
D	sp	104°	90°

8. Which of the following statements concerning the reaction $J \longrightarrow 2K$ is always true?

- A The overall order of the reaction is one.
- B Increasing the concentration of J increases the rate of reaction.
- C Removing K as it is formed helps to increase the rate of reaction.
- D Addition of catalyst increases the rate constant of the reaction.

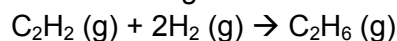
9. Which one of the following statements about $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is correct when the system is at equilibrium?

- A The ratio of the rates of the forward reaction to that of the reverse reaction is equal to the equilibrium constant.
- B The rate constant for the forward reaction equals the rate constant for the reverse reaction.
- C The rates of both the forward and reverse reactions are equal to zero.
- D The rates of the forward and reverse reactions are equal

10. The table below shows the enthalpy change of combustion of ethyne (C_2H_2), hydrogen (H_2) and ethane (C_2H_6).

Substance	$\Delta H_c / \text{kJ mol}^{-1}$
Ethyne	-1300
Hydrogen	-285
thane	-1560

What is the enthalpy change for the following reaction?



- A +310 kJ mol^{-1}
 B +155 kJ mol^{-1}
 C -155 kJ mol^{-1}
 D -310 kJ mol^{-1}
11. Given that the equilibrium constant for the reversible reaction



calculate the K_c for the following reaction: $\text{QR}_2(\text{g}) \rightleftharpoons \frac{1}{2} \text{Q}_2(\text{g}) + 2\text{R}(\text{g})$

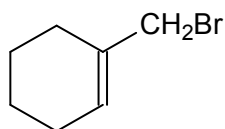
- A 1.8
 B $\frac{1}{1.8}$
 C $\frac{1}{3.6}$
 D $\frac{1}{\sqrt{3.6}}$
12. Values of ionic product for water at different temperatures are given below:

Temperature / $^{\circ}\text{C}$	0	25	100
$K_w / \text{mol}^2 \text{ dm}^{-6}$	1.1×10^{-15}	1.0×10^{-14}	5.1×10^{-13}

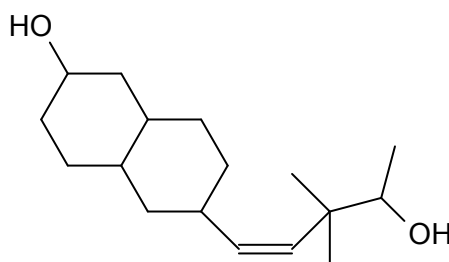
Which of the following statement is correct?

- A The water at 100°C is not neutral.
 B The pH of water at 0°C is 7.
 C The enthalpy change for the dissociation of water is endothermic.
 D Formation of water molecules is favoured at higher temperatures.

13. Which type of reaction does the following compound not undergo?

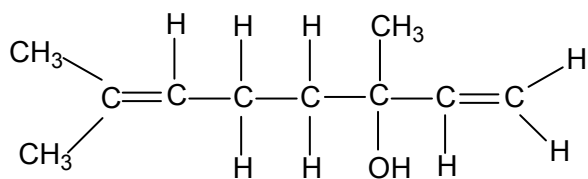


- A Elimination
 B Nucleophilic Substitution
 C Free radical Substitution
 D Electrophilic Addition
14. How many geometrical isomers can compound **T** has when it undergoes reaction with excess concentrated sulfuric acid at 170°C?



Compound **T**

- A 1
 B 2
 C 4
 D 8
15. Deuterium, D, is an isotope of hydrogen.
 Considering only the major product, how many deuterium atoms will be incorporated into a molecule of Linalo-ol when it is reacted with bromine in heavy water, D₂O?



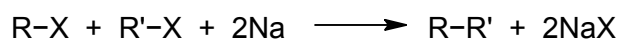
Linalo-ol

- A 0
 B 1
 C 2
 D 4

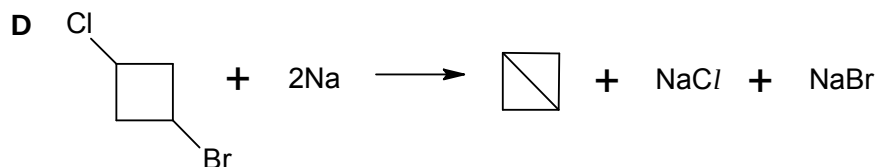
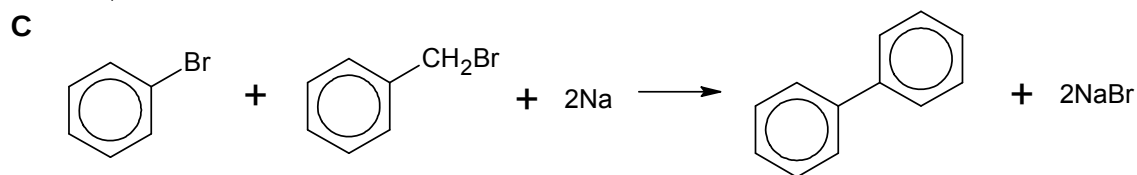
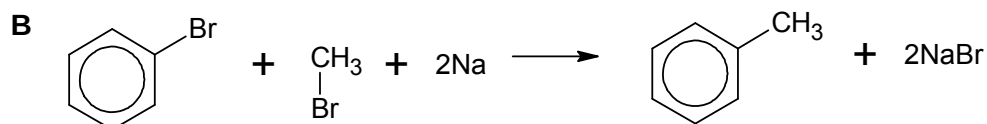
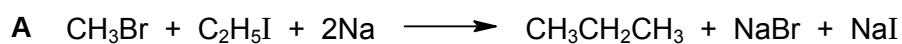
16. Which property of benzene is a result of the delocalisation of electrons in the benzene ring?

- A Substitution in benzene takes place at one particular carbon atom.
- B Addition reactions of benzene take place more easily than substitution.
- C The carbon-carbon bond lengths are between those of C-C bonds and C=C bonds.
- D Benzene is a good conductor of electricity.

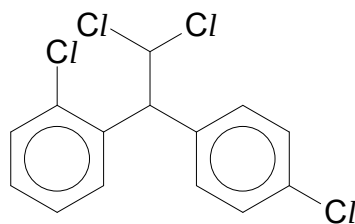
17. In the Wurtz reaction, two halogenoalkanes molecules react with sodium metal to form a new carbon-carbon bond, resulting in the formation of a new alkane:



Which of the following does not show the correct product when the stated reactants are reacted together in a Wurtz reaction?



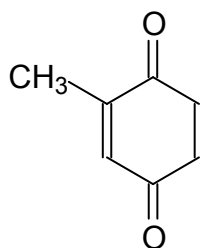
18. *Mitotane* is used as an antineoplastic drug, which helps to inhibit growth and formation of tumours. It has the structure as shown.



Mitotane

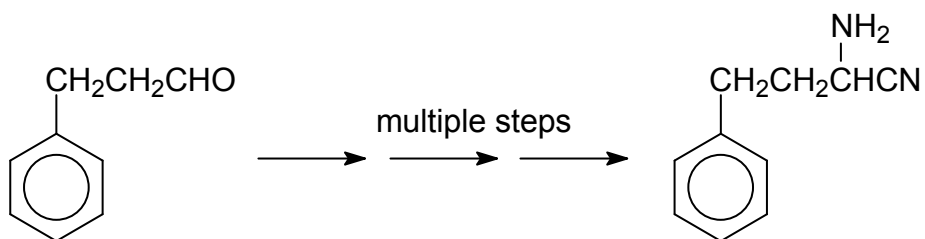
Which of the following statement about Mitotane is true?

- A** It will react with 2 mol of hot ethanolic HCN to produce nitrile-containing products
B It will react with hot aqueous silver nitrate to produce cream precipitate of silver chloride.
C It will react with 4 mol of hot concentrated NH_3 in a sealed tube to produce amine-containing products.
D It will react with ethanolic KOH to form a product that cannot exhibit geometrical isomerism.
19. Compound **U**, $\text{C}_4\text{H}_8\text{O}$ can be oxidised by acidified aqueous potassium manganate(VII) to form $\text{C}_3\text{H}_4\text{O}_4$. Which of the following could be the structural formula of Compound **U**?
- A** $\text{CH}_2\text{CHCH}(\text{OH})\text{CH}_3$
B $\text{CH}_2\text{CHCH}_2\text{CH}_2\text{OH}$
C $\text{CH}_3\text{CHCHCH}_2\text{OH}$
D $\text{CH}_3\text{CHC}(\text{OH})\text{CH}_3$
20. The compound shown below is excreted by the bombardier beetle. How will the compound react with 2,4-DNPH, Tollens' reagent and alkaline aqueous iodine?



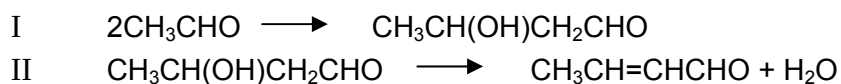
- | | 2,4-DNPH | Tollens' reagent | Alkaline aqueous iodine |
|----------|-----------------|-------------------------|--------------------------------|
| A | Negative | Negative | Negative |
| B | Positive | Negative | Negative |
| C | Positive | Positive | Negative |
| D | Positive | Positive | Positive |

21. The synthesis of a compound X involving multiple steps are shown below:



Which of the statements is correct?

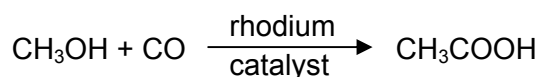
- A The first step is a nucleophilic reaction.
 B The reagent and condition for the first step involves NaCN at 10°C.
 C The synthesis can be achieved in two steps.
 D The last step is an electrophilic reaction.
22. The Russian composer Borodin, was also a research chemist. He discovered a reaction in which two ethanal molecules combine to form a compound commonly known as aldol (reaction I). Aldol forms another compound on heating (reaction II).



Which of the following best describes reactions I and II?

- | I | II |
|----------------|-------------|
| A Addition | Elimination |
| B Addition | Reduction |
| C Elimination | Reduction |
| D Substitution | Elimination |

23. One industrial method of preparing ethanoic acid is the direct carbonylation of methanol using a rhodium catalyst.



Which compound could be expected to produce $\begin{array}{c} \text{COOH} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{COOH} \\ | \\ \text{CH}_2\text{COOH} \end{array}$?

- A** $\begin{array}{c} \text{OH} \\ | \\ \text{H}-\text{C}-\text{COOH} \\ | \\ \text{CH}_2\text{OH} \end{array}$
- B** $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{COOH} \\ | \\ \text{CH}_2\text{COOH} \end{array}$
- C** $\begin{array}{c} \text{OH} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{COOH} \\ | \\ \text{COOH} \end{array}$
- D** $\begin{array}{c} \text{OH} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{COOH} \\ | \\ \text{CH}_2\text{OH} \end{array}$
24. An element **Y**, found in period 3, is a good conductor of electricity in solid and liquid state. It reacts violently with oxygen and water.
- Which of the following statements is correct about element **Y**?
- A** The oxide of **Y** reacts with alkali and not acid.
- B** Element **Y** has a giant metallic structure.
- C** The value of the first ionisation energy of element **Y** is bigger than the value of first ionisation energy of Argon.
- D** The mobile charged carriers of **Y** in solid state are ions and electrons.
25. Which of the follow properties of beryllium compounds is incorrect?
- A** Beryllium oxide undergoes neutralisation with both acids and bases.
- B** Beryllium oxide has a high melting point.
- C** Beryllium chloride reacts with ammonia in a 1:2 ratio.
- D** Beryllium chloride conducts electricity when molten.

For **questions 26 - 30**, one or more of the numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it useful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

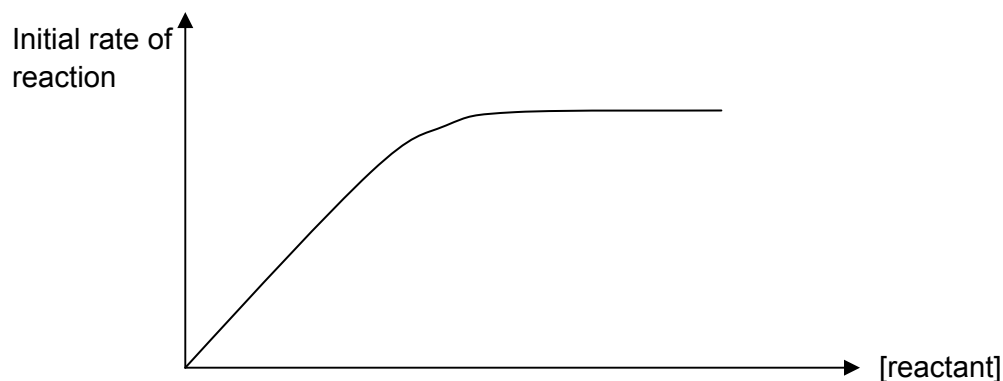
No other combination of statements is to be used as the correct response.

- 26.** Two colourless liquids were mixed together in a beaker and left to stand. The mixture separated into two distinct layers after standing for an hour.

Assuming that the liquids did not undergo any reaction with each other, which pairs of liquids, when mixed, will produce the above observation?

- 1** Ethanal and water
- 2** Cyclohexanol and methanol
- 3** Ethanol and tetrachloromethane

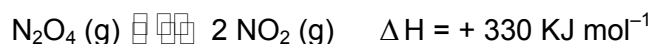
- 27.** The graph shows the results of an investigation of a reaction catalysed by enzymes. In the experiment, the concentration of the reactant is varied while that of the enzyme is kept constant.



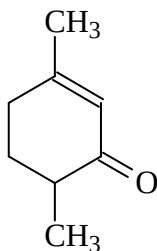
What can be deduced from the graph?

- 1** When [reactant] is high, the rate is independent of [enzyme].
- 2** When [reactant] is high, the rate is independent of [reactant].
- 3** When [reactant] is low, the rate is first order w.r.t. the reactant.

28. Which of the following statement(s) is/are correct for the following equilibrium?

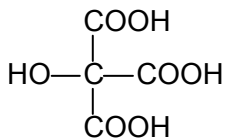


- 1 The units of K_c is mol dm^{-3}
 - 2 Position of equilibrium will shifts right when there is a decrease in pressure.
 - 3 K_c increases with the increase in temperature.
29. Which of the following statement(s) is/are true piperitone?



piperitone

- 1 When 1 mol of piperitone reacts with cold potassium manganate(VII), the product formed will produce 1 mol of hydrogen gas when reacted with Na metal.
 - 2 It reacts with acidified potassium dichromate (VII).
 - 3 It reacts with Fehling's reagent to give a reddish brown precipitate.
30. Citric acid, which causes the sharp taste of lemon juice, has the following structure.



What of the following statement(s) about reactions involving citric acid is/are true?

- 1 2 mol of H_2 is produced when 1 mol of citric acid reacts with Na (s)
- 2 3 mol of PCl_5 (s) is required to react with 1 mol of citric acid
- 3 4 mol of NaOH (aq) is required to react with 1 mol of citric acid

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